

Lab 7 CheckIn – Oliver Wuttke (WUTT0019)

Question 1 – Progress

What did you accomplish this week? Any issues slowing down progress?

This week I did all my part of the code for the A1 part of the capstone project. I also did the lab as per usual. The main issue slowing down progress was learning how to use the Open Electricity SDK, but once I played around with all the examples I was able to do my part smoothly.

Question 2 – Plans

What are your goals for next week? How does this match against your overall project plan?

To do my report and video for the A1 assessment. This matches up good with the project plan since the code needed to be done first.

Question 3 – Study

List and discuss the three main points of material covered this week.

1. MLP Models – In the lab we looked at using MLP models to forecast and compared them against the ARIMA forecasting models. We looked at a variety of methods for improving MLP model accuracy from feature scaling to lagging features.
2. 1st Order Logic – Continued looking at 1st Order Logic in the lecture, mainly covering logical operators and atomic sentences.
3. N/A

Question 4 – Insights

What was the single most confusing concept (“muddiest point”) for you in this week’s material, and what specific question do you have about it?

My biggest struggle was re-learning the PyTorch basics for question 5 in the lab. Had to just go over some old notebooks from the Machine Learning and Neural Networks topic as well as re-read some of the PyTorch documentation.

Question 5 – Linkage

How could one key idea from this week’s material be directly applied to your project?

I'm thinking of doing forecasting for one of the A2 models (forecasting demand) and using a neural network to capture complex patterns may very well be a good idea. But I'm particularly interested in maybe using a Bayesian neural network instead since the stakeholders for the project will likely want uncertainty quantified.